Fénix (I) — Engineering Brief for a Circuit-Level Software Emulation

Handoff document for an implementing agent. Compiled 2026-11-03 from designer documentation (Bert Vermeulen, dutchsynth.eu), the Sound On Sound review (Feb 1999), the Fénix II PP manual (shared conventions), and owner reports. Every claim is tagged with a confidence level; the implementer must not silently upgrade a 'moderate' or 'low' claim into a design assumption.

0. How to use this brief

- **Target:** a software instrument that reproduces the Fénix (1997–2000, 75 units, Marc Paping + Bert Vermeulen) at the level of individual circuit blocks: expo converters, OTA state-variable filters, a 5-pole ladder, an LFSR noise source, four-quadrant multipliers, RC/op-amp envelope shapers, and a free banana patch graph with feedback.
- **Hard constraint:** no schematics of any Fénix have ever been published. 'Circuit-level' therefore means *topology-level* modelling driven by (a) the designer's own part list and trimmer descriptions, (b) measured/reviewed behaviour, and (c) the well-known canonical circuits those parts imply. Sections 4–5 state the inferred topology for each block and the evidence for it.
- **Reading the tags:** [HIGH] = stated by the designer or two independent sources agree. [MOD] = one source, or strong inference from the parts list. [LOW] = plausible reconstruction; must be exposed as a tunable and verified against hardware recordings. [UNK] = unknown; a default must be chosen and documented.
- **Deliverable expectations for the implementing agent:** (1) a patchable modular engine with every module below, (2) a measurement harness (section 7) that runs on every build, (3) a 'topology parameters' file that collects every [MOD]/[LOW] constant so it can be retuned without touching DSP code.

1. Identity, scope and what this is NOT

- Fénix (I) is a semi-modular with **31 modules**, **63 pots, 3 switches, 158 banana sockets** (designer's count) [HIGH]. No internal normalling: nothing sounds until patched [HIGH].
- It is **not** the Fénix II (2010, different and ~3x larger design), nor the Fénix III (a sequencer), nor the Fénix II PP / IId (Patch Point and kit versions of the II), nor the TINRS Fenix IV (unrelated circuitry). Only the shared electrical conventions (section 2) carry over from those documents; do not import II-series module behaviour into this emulation.
- Design philosophy stated by the designer: discrete transistors and commodity ICs (TL074-class op-amps, CA3080/CA3280/LM13700 OTAs, CD4000 CMOS), no Curtis/SSM synth ICs [HIGH]. Later units had VCF3 retuned to be 'more Moog-ish' after owner feedback [MOD, SOS/Chris Whitten]. The emulation should expose an 'early/late VCF3 voicing' switch (section 4.5).

2. System-level electrical contract (applies to every module)

Quantity	Value	Conf.	Source / note
Pitch & cutoff scaling	1 V/oct on every 'CV1' input; CV1 inputs in mixers are calibrated unity gain	HIGH	SOS, designer, II PP manual
CV2 inputs	Panel attenuator. Two laws: 0→max (unipolar) and –/0/+ (bipolar, zero at 12 o'clock; CCW inverts)	HIGH	Designer spec page
Audio level	4 Vpp nominal (±2 V). Reviewer measured 'hot enough' to hear by touching; excellent SNR	HIGH	SOS, Matrixsynth, JonDent
CV ranges	–4…+4 V (bipolar sources) or 0…+8 V (EGs, positive LFO outs)	HIGH	SOS, designer, JonDent
Gate convention	Positive-going; EGs/triggers accept 'a wide range of voltages'. Adopt II PP thresholds: high > +5 V, low < +1 V, out 0 / >+7 V	MOD	SOS; thresholds from II PP manual
Absolute swing	Rails ±15 V (service page: ±15 V test points). Treat ±13.5 V as op-amp clip (TL07x on ±15 V)	HIGH/MOD	Service page

Quantity	Value	Conf.	Source / note
Output buffering	Outputs are buffered, short-circuit proof; drive ≥5 inputs / ≥1 kΩ total without droop	HIGH	Designer FAQ
Input impedance	High (op-amp inputs). Unbuffered external MIDI-CV outputs droop when driving several Fénix CV1s — i.e. inputs are ~10–100 kΩ each	MOD	Service page, calibration page
Banana stacking	Stacking at an **input** = fan-out (fine, up to ~6 deep). Stacking two **outputs** together is NOT linear summing: outputs fight; 'at best unpredictable'	HIGH	SOS 'Right Connections' box
Colour code	Black = input, Green = audio out, Blue = CV out	HIGH	SOS, PC
Thermal	Each VCO and VCF core heated to 60 °C; bi-colour LED red→green at temperature. Consequence: emulate NO long-term drift; only a short warm-up (optional cosmetic)	HIGH	SOS
Rear I/O	6 front bananas hard-wired to 6 rear 3.5 mm jacks (2 CV, 2 Gate, 2 Aux); rear ground post; 2 front ¼″ audio inputs; ¼″ TRS stereo out (L=tip, R=ring)	HIGH	SOS, designer
Power	External 17 V/17 V @ 880 mA; internal ±15 V regulators at rear panel	HIGH	SOS, service page

Modelling implication for stacked outputs [MOD]: each op-amp output has a small series/output resistance and current limit. Two outputs tied together resolve to a current-limited average, not a sum. Model as: node voltage = weighted mean of connected outputs (weights = 1/R_out), then clip each driver's current to ~±20 mA (TL07x class). This reproduces 'unpredictable' behaviour deterministically and is cheap. Expose R_out (default 100 Ω) and I_limit as parameters.

3. Parts list → circuit-family inference

The designer publishes the uncommon parts and the common parts. This table is the primary evidence for the topology choices in section 4. Quantities in italics on the designer's page mean '1 or 2 used in the whole instrument'.

Part	Function in audio circuits	Where it almost certainly lives in the Fénix	Conf.
TL074 / TL072	JFET quad/dual op-amp	Everywhere: mixers, buffers, integrators, EG shapers	HIGH
CA3280 (dual OTA with linearising diodes)	Low-distortion VC gain element	VCF1/VCF2 state-variable integrators; VCAs	MOD
LM13700, CA3080 (1–2 units each)	OTA	VCA(s), ring-mod cores, or S&H;/slew elements	MOD
CA3046 (5-transistor array)	Matched NPN pairs; one pair used as heater + sensor	Expo converters of VCOs and VCFs, plus the 60 °C on-chip heater (same scheme as Doepfer A-110)	HIGH (heater), MOD (expo pair)
BF245 (N-JFET)	Reset switch / S&H; switch / low-leakage buffer	VCO sawtooth-core discharge switch; S&H; hold switch	MOD
BC547/BC550/BC557	Small-signal BJTs	Expo converters, ladder/ cascode stages, level shifters	HIGH
CD4006 (18-stage shift register) + CD4070 (XOR)	LFSR	Digital noise generator (designer: 'cycle period ≈ one year', clocked by a VCO)	HIGH
CD4001 (NOR)	Logic	Gate conditioning, LFO reset, comparator latch, EG3 delayed-gate	MOD
LM311	Comparator	Comparator module; PWM comparators; gate detectors in ESP modules	HIGH
CA3140 (1–2)	MOSFET-input op-amp	S&H; hold buffer (low bias current) and/or slew limiter	MOD
MC34084 (1–2; subs TLE2074/TL054)	Fast quad op-amp	VCO cores (fast reset) or VCF3	MOD
Radiohm CIP160C pots; laws A=lin, B=log, C=anti-log (reverse log), S=S-curve	Panel controls	See per-module taper list in section 4 — tapers are part of the feel and must be modelled	HIGH

What the parts list rules out: no CEM/SSM/AS chips, no vactrols (those appear only in Fénix II), no BBDs, no digital control. Everything in the Fénix I is continuous-time analog except the LFSR noise clock domain.

4. Module-by-module specification

Format per module: panel (control, pot law), I/O, documented behaviour, inferred topology, recommended DSP model, parameters to expose. Pot laws are from the designer's spec page [HIGH] unless noted. 'lin' = A, 'log' = B (audio taper), 'C' = reverse-log, 'S' = S-curve.

4.1 VCO1 (full-featured) and VCO2/VCO3 (basic)

	VCO1	VCO2 / VCO3
Controls (all 10k lin)	Frequency, Tune, CV2 amount, PW, PWM depth, Sync level (soft→hard continuous per II-PP doc; SOS says hard/soft/none)	Frequency, Tune, CV2 amount
CV inputs	CV1 (1 V/oct), CV2, linear FM, PWM, Sync in, Sync level CV, Sub-phase CV	CV1, CV2, linear FM, PWM
Audio outputs	Sine, Saw, Pulse, Sub1 (square, –1 oct), Sub2 (square, –2 oct)	Saw, Pulse
Range	Sub-audio to above audio ('VCOs go as low as LFO rates')	same

Inferred topology [MOD]: classic discrete sawtooth core — expo converter (matched BJT pair on a heated CA3046, 1 V/oct trim, high-frequency compensation trim, initial-frequency trim, tempco trim), integrator charging a capacitor, LM311 comparator, JFET (BF245) reset. Four trimmers per VCO are documented [HIGH], matching exactly this structure. VCO1 has three further trimmers for triangle and sine shape [HIGH], implying saw→triangle folding (full-wave rectify + offset) and triangle→sine via a differential-pair (or diode) soft-clipper — the standard discrete sine-shaper whose residual THD is ~1–3 %. Sub-oscillators are flip-flop dividers (÷2, ÷4) of the comparator; the **sub-phase CV** shifts both subs by up to ±1 period of the fundamental [HIGH, SOS], which implies the dividers are retriggered from a phase-delayed copy of the reset pulse — a voltage-controlled one-shot delay (0…T) in front of the divider chain. Side effect noted in the II-PP manual for the same feature: pitch of the sub outputs shifts while the phase is being modulated (d(phase)/dt = frequency offset). Reproduce that.

Sync: hard sync = reset core on rising edge; soft sync = reset only if the slave is within a phase window near its own end-of-cycle; the window width is the 'sync level' control and CV [HIGH for II PP, MOD for Fénix I which SOS describes as a three-way hard/soft/none selection plus amount]. Implement: window parameter 0 (off) → 1 (hard), continuously variable; frequency-ratio dependent misses fall out naturally. Sync only responds to positive edges [HIGH].

Linear FM: through-zero? [UNK]. The II PP manual says its FM inputs are AC-only and detune the perceived pitch with modulation index (not through-zero compensated). Default to non-through-zero linear FM applied as an additive current into the integrator, allowing the core to stall at zero (not reverse). Expose a through-zero option off by default.

PWM: comparator on the saw/tri against PW + PWM·CV. Pulse disappears when the threshold exits the ramp [HIGH, II PP]. Pulse width window: ±4 V CV swings full range at PW=12 o'clock.

DSP model: run the oscillator core at ≥4x oversampling (or use a PolyBLEP/DPW saw + BLAMP triangle with a sync-aware reset; sine via the measured shaper transfer curve). Keep the expo converter as a function f = f0·2^(V/1) with a bulk-resistance high-frequency droop term r·I (this is what the 'high frequency' trimmer corrects) and expose the residual as 'HF tracking error' (default 0 — the instrument is calibrated). Model comparator reset delay (~100–300 ns in hardware → saw amplitude shrinks slightly at high f): expose as 'reset time'. Heated core → no thermal drift.

Parameters to expose: sine shaper THD/asymmetry; triangle fold corner; reset time; sync window curve; sub-phase delay law (linear in CV); FM gain; saw/pulse amplitudes (nominally ±2 V each).

4.2 LFO1, LFO2, LFO3

	LFO1	LFO2	LFO3
Controls	Freq, CV2, Symmetry/PW (10k lin, D-shaft)	Freq, CV2, Symmetry/PW (10k lin)	Freq (10k log)
Inputs	CV2, Trigger (reset/sync)	CV2, Gate (on/off: runs while high)	none
Outputs	Triangle (positive-going 0…+8 V), Pulse	Triangle (bipolar ±4 V), double-frequency triangle (positive), Pulse	Sine, Triangle, Pulse
Range	VC, 'reaches audible spectrum'	VC	manual only

Inferred topology [MOD]: LFO1/2 are triangle/square relaxation oscillators (op-amp integrator + Schmitt comparator) with symmetry set by splitting the charge/discharge currents (the II PP documents a 3…97 % duty range for the same design family — adopt). Exponential VC via a small expo pair (not heated). The double-frequency triangle of LFO2 is a full-wave rectification of the bipolar triangle (|x|, rescaled to 0…+8 V). LFO3's sine is a triangle fed through a diode shaper. Reset =

force integrator to one rail (trigger) ; LFO2's gate input likely holds the integrator (run/hold) rather than resetting. Expose both behaviours.

DSP: integrator-based so that symmetry, CV-rate changes and resets behave like hardware (phase continuity, slope change mid-cycle). Audio-rate operation should alias-protect the pulse output (BLEP) since users will patch LFO1 as a low oscillator.

4.3 VCF1 and VCF2 — 2-pole state-variable (12 dB/oct), simultaneous LP/BP/HP

Controls: Frequency, Resonance, CV2 (all 10k lin). Inputs: CV1 (1 V/oct), CV2, Audio (single input). Outputs: LP, BP, HP. [HIGH]

Topology [HIGH that it is an SVF; MOD on element]: two OTA-integrators (CA3280 is the obvious candidate — dual OTA with linearising diodes, used by the designer in the Synton 3000) plus a summing op-amp; resonance = damping feedback from BP. Calibration page: 1 V/oct trim, initial frequency trim, resonance trim set so that the LP output level stays flat when swept at max resonance ('there could be a slight dip') [HIGH]. That flat-at-self-oscillation behaviour means the output is not a simple raw SVF — amplitude is bounded (OTA tanh saturation). Reviewer: multimode filters have 'great character'; the Notch spectrogram on the designer's site is LP+HP summed.

DSP: zero-delay-feedback (TPT) SVF with the integrator inputs passed through tanh(x/Vt_eff) to mimic OTA input clipping (CA3280 with diodes: effective Vt larger → cleaner; expose Vt_eff default ~60–120 mV referred after the input attenuation). Resonance to self-oscillation and beyond; self-oscillation amplitude limited by the tanh. Run at 2x oversampling minimum. Cutoff via 1 V/oct; CV2 bipolar attenuverter.

Parameters: Vt_eff per integrator; resonance-to-damping law (so that 'max' is just beyond self-oscillation, per the calibration note); LED overload threshold (cosmetic).

4.4 VCF3 — 'ladder-type' low-pass, 2-pole tap and 5-pole output, with 3-input mixer

Controls: Frequency, Resonance, CV2 (10k lin) and three audio input levels (10k **log**). Inputs: CV1, CV2, Audio 1/2/3. Outputs: '12 dB' (2-pole tap) and '24 dB' which the designer states is **actually 5-pole**; SOS states five poles giving a theoretical 30 dB/oct. [HIGH]

Topology [MOD]: the designer calls it a ladder. With a CA3046 heater (VCF3 has a dedicated temperature trimmer, measured at pin 7 of IC2 = –0.55 V [HIGH]) and the BJT/JFET parts list, the most probable structure is a transistor ladder (Moog-type) of **five** RC stages driven by an expo current source, with a tap after stage 2 and feedback taken from stage 5. Four trimmers: 1 V/oct, initial frequency, resonance, log-converter temperature [HIGH]. Calibration note: unlike VCF1/2, the output level is **not** stable when swept at high resonance [HIGH] — characteristic of a ladder where the resonant peak amplitude depends on the drive and the input mixer gain ('filter behaviour is partly determined by the input level setting' [HIGH]). Later production had VCF3 'tweaked to be more Moog-ish' [MOD].

DSP: 5-stage nonlinear ladder (Huovilainen-style, each stage $y' = \omega \cdot (\tanh(x) - \tanh(y))$ with feedback $k \cdot \tanh(y5)$ subtracted at the input, 2x–4x oversampling, zero-delay solution via Newton or the usual linearised correction. Resonance range: self-oscillation reachable near max (calibration: trimmer set so it 'just starts to oscillate' at 10 o'clock with no CV). The two voicings ('early' vs 'late/Moog-ish') should be modelled as (a) feedback tap/compensation differences and (b) input drive level — expose both; default to 'late'. The 3-input mixer is log-taper pots into the ladder input: pot law affects drive and therefore saturation — model exactly. Passband gain drops with resonance (classic ladder bass loss) unless the designer compensated; [UNK] — expose compensation 0…1, default 0.

4.5 Envelope generators

	EG1	EG2	EG3
Stages	A–Peak(hold)–D–S–R ('APDSR') + second decay-only 'Accent' envelope	A–Peak–(D or R, switch) + second decay-only envelope; Mix out = 50/50 sum	Delay-then-AD (near-instant attack), plus delayed-gate outputs
Pots	Attack, tPeak: 2M2 log. Decay: 10k C (reverse-log). Sustain: 10k log. Release, Accent: 10k C	Attack, tPeak: 2M2 log. Decay/Release: 10k C. tDecay (2nd env): 2M2 log	Delay, Decay: 2M2 log
CV inputs	Gate, Decay time, Peak level, Release time, Accent time, Accent level	Gate, Decay/Release time, Accent level	Gate
Outputs	CV out, Accent out	CV out, Accent out, Mix out	CV out, delayed gate (+), delayed gate (–)

Inferred topology [MOD]: the 2M2 log pots for Attack/Peak and 10k reverse-log pots for Decay/Release are a strong tell: attack/hold stages are simple RC charge via a pot (exponential-toward-target curve; 2M2·C gives tens of seconds), while decay/release use a 10k pot that sets a *current* into an OTA or transistor discharge — i.e. voltage-controllable (the CV inputs on D and R exist) — giving exponential decays whose time constant is CV-steerable. SOS: A, D, R of EG1 'can all independently be voltage-controlled'. Designer FAQ (II series, same philosophy): envelope controls are optimised for very short times; times from <1 ms to >50 s. The Peak-hold stage is a monostable (CD4001/comparator) that holds the output at max before decay — reproduce as a timer, not a curve. 'Accent' second envelope: instantaneous attack, CV-controlled decay and CV-controlled attack level (SOS) — i.e. a capacitor charged to a CV level and discharged through a VC current sink. EG3's delayed gate: timer from gate-on, then a gate of both polarities.

DSP: stage-based one-pole integrators with exponential (RC) segments; attack curve = 1–e^{–t/τ} towards a target above full-scale (so the attack ends at ~63–90 % of the way, like real RC attacks — expose the overshoot target, default 1.4x); decay/release = e^{–t/τ} with τ from the reverse-log pot and CV (1 V ≈ ×2 time change, [LOW] — II PP says higher CV = longer time). Output 0…+8 V. Re-trigger behaviour from current level (no reset to zero) [MOD].

4.6 VCAs (×4), Mixers, CV mixers, Portamento, Slew, S&H;, Comparator

Module	Panel / I/O [HIGH]	Topology & model [MOD unless noted]
VCA1–4	In, CV, Out. No controls. 'Suitable for audio and CV'	Linear OTA VCA (LM13700/CA3080 or half a CA3280), 0→unity, DC-coupled. CV ≤ 0 → closed. Model: gain = clamp(CV/8, 0, 1) with OTA tanh on the signal path (Vt_eff ~ 60 mV after internal attenuation); mild CV feed-through (expose, default –60 dB).
Mixer1	A (10k lin), B (–/+ attenuverter), X unity. Out	Op-amp summing. B inverting attenuverter. DC-coupled (usable for CVs; SOS: has a 1 V/oct input). Clip at rails.
Mixer2, Mixer3	A–B balance (10k **S-curve**), X unity	Crossfade pot with S-law: model the S taper explicitly (constant-power-ish feel). DC-coupled.
CV Mix1	Gain (100k lin), Offset (10k lin); In, X unity	Attenuate In + offset + unity X; offset range about ±4 V [LOW].
CV Mix2	Octave switch –1/0/+1/+2 V; CV2 (100k lin); CV In unity	Calibrated 1.00 V steps (designer uses it as calibration source) — exact.
Portamento	Time1, Time2 (2M2 log); In, Enable (gate); Out1, Out2	ONE input, two independent RC lag outputs; Enable gate defeats (bypasses) both. Unity-gain lag (designer: slew limiter is NOT unity, portamento is).
Slew limiter	Rise, Fall (2M2 log); In, Out	Separate rise/fall rate limiting (current-limited integrator: diode-steered). Designer: 'no defined unity gain' — a small gain error (expose, default ±2 %). Usable as AR envelope / envelope follower.
S&H;	In, Trigger, Out	JFET switch + hold cap + CA3140 buffer. Droop: II PP manual warns the stored voltage 'may gradually decrease' — model droop ~ tens of mV/s (expose); sample on rising edge; aperture glitch optional.
Comparator	+, – inputs; momentary push button; Gate out	LM311: out high when (+) > (–); push-button forces gate high. Small hysteresis (expose, default 20 mV). Audio-rate capable → PWM/square conversion.

4.7 Ring modulators

RM1: inputs A, B, 'am' (AM/cleanfeed CV mixed into B); outputs DC-coupled and AC-coupled (AC version 'eliminates offset clicks') [HIGH]. SOS behaviour: A×B gives sum/difference *plus* both carriers (so it is not a perfect balanced modulator — it has carrier leak); inputs 1 and 3 give the same at lower amplitude; a square on input 3 gates the modulation on/off, letting A through unmodulated in the 'off' half with a little RM bleed. **Interpretation [MOD]:** four-quadrant multiplier (OTA-based, CA3080/LM13700) where the 'am' input is summed with B so that B_eff = B + am; output = A·B_eff/4 + leak_A·A + leak_B·B. SOS's observations are exactly reproduced by B_eff = B + am with am DC offset acting as 'cleanfeed' of A. **RM2**: plain A, B → DC-coupled out. Designer has a dedicated ring-mod calibration page (carrier null trims) [HIGH] — so leakage is a trimmed residual: expose leak_A, leak_B (default –40 dB) and the ×1/4 scaling (II PP states A·B/4 for the same family).

4.8 Noise module — analog, digital (VC bandwidth), VLF random

Controls: Timbre, CV2 (10k lin); Analog level, Digital level (10k log); VLF level (50k log). Input: Timbre CV. Outputs: Analog, Digital, VLF. [HIGH]

Digital noise [HIGH on structure]: designer: 'cycle period of approx. one year; the clock is a VCO, so you can control the bandwidth using the control or a CV.' Parts: CD4006 (18-stage shift register) + CD4070 (XOR) → maximal-length LFSR. A one-year period at an audio-rate clock implies ~2^41 … 2^45 states, i.e. two or three 4006s cascaded (e.g. 43–46 bits) [MOD]; the exact taps are unknown [UNK] — any maximal LFSR ≥ 40 bits clocked at a VC rate is perceptually equivalent. Output is the raw 1-bit stream (two-level) through a level pot; the **timbre** control sweeps the clock from sub-audio (random pulse trains / 'tonal growl') to >50 kHz ('blue noise', SOS). Model: LFSR clocked by an expo VC clock f = f0·2^(CV); output ±2 V bit-stream, band-limited by the sample rate — render with a zero-order-hold at oversampled rate and decimate so the characteristic sin(x)/x spectrum of clocked 1-bit noise appears.

Analog noise [MOD]: reverse-biased BJT junction amplified; SOS suspects pink-ish ('roaring'). Model: white Gaussian with a gentle –3 dB/oct tilt filter (expose tilt 0…–3 dB/oct, default –1.5). **VLF [HIGH]:** slow random derived from the analog noise (II PP: 'derived from the analog noise source', max a few Hz) — model as noise through a 2-pole low-pass at ~1–3 Hz, bipolar ±4 V.

4.9 External signal processors (Audio-in 1 and 2) and Stereo output

Audio-in 1: Gain (100k lin); ¼″ jack → Audio out, Level CV (envelope follower), Trigger (fires on abrupt positive level change). **Audio-in 2**: Gain (100k lin), Threshold (10k lin) → Audio out, Envelope, Gate (comparator on envelope vs threshold). SOS: 'not entirely reliable' detection; one gain control mislabelled. Gain range: II PP states –6 dB…+20 dB for its input module — adopt. Model: op-amp gain stage; envelope follower = precision rectifier + fast-attack/slow-release RC (~2 ms / ~50 ms, [LOW]); trigger = one-shot on d(env)/dt > threshold; gate = comparator with hysteresis. **Stereo out**: Tip (L) and Ring (R) attenuators (2×220k lin), drives headphones; model as a clean buffer with clip at ±13.5 V.

5. Cross-cutting circuit modelling rules

- **Unified signal rate:** there is no CV/audio distinction in hardware. Every connection is a sample-rate signal; no control-rate decimation anywhere. Audio-rate modulation of filters, VCAs, S&H; and ring mods is the instrument's main attraction (SOS).
- **Feedback patching:** any output may be patched to any input, including self-feedback. Use a graph scheduler that detects cycles and inserts a one-sample delay at cycle-break edges, with the engine oversampled (2x–4x) so the delay is ≤ ~6 µs — comparable to op-amp slew/propagation in hardware. Document which edge got the delay.
- **Saturation everywhere:** op-amp stages clip at ±13.5 V (soft knee ≤ 0.5 V); OTA stages saturate as tanh with Vt_eff; integrators clip at rails. The instrument's hot 4 Vpp levels mean mixers can be driven into clipping on purpose (designer: 'the sounds can be distorted at various stages of a patch').
- **Calibration-perfect by default:** the instrument is trimmed and heated; default all tracking errors to zero. Provide a 'vintage' macro that reintroduces small 1 V/oct errors (≤ 0.3 %) and HF droop per VCO — off by default.
- **Pot laws are behaviour:** implement the exact taper per control (lin, log, reverse-log, S). For the 2M2 log time pots use the standard 'B' audio taper (10 % at mid-rotation); for the 10k C reverse-log pots the mirror image. S-curve: use a smoothstep-like mapping (expose).
- **Input loading:** fan-out is free; stacked outputs resolve by the current-limited average rule in section 2.
- **DC coupling:** everything is DC-coupled except RM1's AC output and the ESP audio paths (assume 20 Hz HPF [LOW]).
- **Noise floor:** add ~–100 dBFS white noise per op-amp stage (cosmetic; expose).

6. Plugin architecture recommendation

- **Core:** C++ (JUCE or CLAP/VST3 via a thin wrapper), or Faust for the DSP blocks compiled to C++ — the module set maps cleanly to Faust's block-diagram semantics, and the user's own toolchain (Faust, gen~, Numba) can be used for reference models and A/B testing.
- **Engine:** a directed graph of 31 module nodes with named ports matching the panel legends (keep the designer's names: CV1, CV2, X, tPeak, Accent, am, VLF, etc.). Patch cables are edges; stacking is multi-edges. Per-block sample processing at the oversampled rate with per-module state; topological order with cycle breaking.
- **Oversampling:** 2x for the whole graph, 4x local for VCO cores and VCF3; polyphase half-band decimation on the way out.
- **UI:** the physical layout matters for usability — reproduce the 11-strip layout and the colour-coded jacks; show the heater LED (red→green after ~60 s cosmetic warm-up).
- **Rear-panel I/O:** expose the six rear jacks as plugin inputs (CV/Gate/Aux pairs) plus a MIDI-to-CV adaptor that produces 1 V/oct CV and a positive gate — note the real instrument has no MIDI.
- **Presets:** store the patch graph + knob states; include the SOS review patches and JonDent's VCO1→VCF3 stereo-filter test patch as reference presets.
- **Tunable constants file:** every [MOD]/[LOW]/[UNK] value above (Vt_eff per stage, ladder compensation, RM leakage, S&H; droop, EG overshoot target, LFSR length/taps, noise tilt, output R/I-limit) lives in one JSON so retuning against hardware recordings does not require recompiling.

7. Verification plan (must ship with the plugin)

No hardware is available for null tests, so verification is by measurement against documented behaviour and published audio. The harness should run headless on every build and emit a report.

Test	Method	Pass criterion / reference
VCO 1 V/oct tracking	Sweep CV1 –2…+6 V in 1 V steps, FFT peak	±0.1 cent per octave at default (calibrated); 'vintage' macro introduces documented errors only
Sine shaper THD	1 kHz sine out, THD+N	1–3 % with odd-harmonic dominance (discrete shaper); expose and document the chosen value
Sub-phase	Sine LFO into Sub-phase CV at 0.5 Hz	Sub outputs show vibrato (frequency offset = dϕ/dt); zero DC pitch error at rest

Test	Method	Pass criterion / reference
Sync	Slave 220 Hz, master 110…880 Hz sweep, window 0→1	Hard: classic sync spectra; soft: intermittent resets whose rate depends on frequency ratio
VCF1/2 response	White noise / sine sweep at Res = 0 and max	12 dB/oct LP/HP/BP; LP level flat (≤1 dB dip) across sweep at max Res — matches calibration note
VCF3 slope	Sine sweep, both outputs	2-pole tap ≈ 12 dB/oct; main output 30 dB/oct asymptote; level varies with resonance (not flat) — matches calibration note; self-osc at Res max
Aliasing	VCO saw at 5 kHz, FFT above Nyquist/2 of folded components	Aliased components < –80 dBFS at 48 kHz host rate
Ring mod	Two sines 400/300 Hz	Sum/difference dominant; carrier leak at the configured –40 dB; 'am' square gates modulation on/off
Digital noise	Timbre sweep; spectrum	sin(x)/x-shaped spectrum following clock; period > 1e9 samples; two-level output
EG timing	Gate step; measure 10–90 % times across pot range	<1 ms … >50 s; attack RC-shaped; D/R exponential; peak-hold is flat
Feedback stability	VCF1 BP → VCA → VCF1 In loop	No NaN/inf; bounded by tanh; one-sample delay edge logged
Stacked outputs	Two sines at one input vs two outputs stacked	Input stacking = clean sum; output stacking = average + current-limit distortion
CPU / latency	Full 31-module patch at 48 kHz, 64-sample block	Report; target < 15 % of one core; latency = decimation filter only
Listening reference	JonDent 'Fenix 1 stereo-filter modulation' WAV (VCO1 sine/sq/ramp + subs → VCF3 12/24 outputs); SOS review descriptions	Spectral envelope and resonance behaviour qualitatively match; document deviations

8. Open unknowns and how to close them

- **Board overlays (11 PDFs on dutchsynth.eu/ServiceF1.html: vco1, VCO2, VCF1, VCF2, AD, ADSR, Mix, PRM, LFO, Int, Div):** component overlays with designators and IC positions. Could not be retrieved in this session (host not reachable from the sandbox). They will settle: which OTA is in VCF1/2, whether VCF3 is BJT ladder or OTA cascade, the LFSR length (count the 4006s), and the EG discharge element. **Action:** download in a browser and hand them to the implementing agent before DSP work on VCF3 and the noise module.
- **Owner recordings:** any direct-out recordings of individual modules (raw saw, sine, VCF3 sweeps at known resonance) would allow actual spectral fitting of Vt_eff, ladder compensation and shaper THD. JonDent's 2015 WAV is the only public one found.
- **Designer contact:** Bert Vermeulen answers questions by e-mail (dutchsynth.eu/Contact.html); SOS reported unlimited support as a selling point. Specific questions worth asking: VCF3 topology and the 'Moog-ish' revision; RM1 cleanfeed structure; LFSR length; EG D/R current-control element.
- **Fénix II PP manual** (public PDF) documents the same designer's later implementations of several blocks (sync window, PWM range, VCA/RM scaling, S&H; droop, EG CV polarity). Use it as the tiebreaker for [UNK] behaviours, but flag every borrowed value.

9. Sources

- Sound On Sound, 'Synton Fénix' review, Gordon Reid, Feb 1999 — https://www.soundonsound.com/reviews/synton-fenix
- DutchSynth (Bert Vermeulen) — Fénix I spec & pot laws: https://dutchsynth.eu/SpecsF1.html
- DutchSynth — Servicing (parts, fuses, PSU, board layout PDFs): https://dutchsynth.eu/ServiceF1.html
- DutchSynth — VCO calibration: https://dutchsynth.eu/Calibrating-VCO-F1.html ; VCF calibration: https://dutchsynth.eu/Calibrating-VCF--F1.html ; VCA, RM, CV1 pages linked from the same menu
- DutchSynth — Filters (spectrograms): https://dutchsynth.eu/FilterF1.html ; Spare parts: https://dutchsynth.eu/Spareparts.html
- Patch Point — Fénix II PP manual (shared conventions only): https://cdn.shopify.com/s/files/1/0626/7600/9207/files/Fenix_Manual.pdf?v=1731589205
- Perfect Circuit — 'Deep Dive with the Fénix': https://www.perfectcircuit.com/signal/synton-fenix-deep-dive

- Matrixsynth 2006 listing (levels, power, build notes): https://www.matrixsynth.com/2006/10/synton-fenix-modular.html
- JonDent — Fénix 1 filter modulation test & WAV (serial #18, built 1998-02-10): https://djjondent.blogspot.com/2015/01/synton-fenix-1-filter-modulation.html
- Wikipedia — Synton Fenix (production history): https://en.wikipedia.org/wiki/Synton_Fenix